

CYBER SECURITY INTERNSHIP

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TASK 1

**SCAN YOUR LOCAL NETWORK FOR
OPEN PORTS**

TOOLS USED: NMAP & WIRESHARK

1. INTRODUCTION

Network scanning is an important aspect of cybersecurity reconnaissance and network security assessment, as it helps identify active hosts, open ports and network services that are accessible within a network. Understanding these exposed services is essential for assessing a network's attack surface and identifying areas that may require additional security controls.

As part of this task,

Nmap 7.98 was used to perform a TCP SYN scan on the local network 192.168.88.0/24. The scan was conducted to discover active devices, identify open TCP ports, determine the commonly associated services and assess the potential security risks related to those exposed services. The identified services were further researched to understand their purpose and possible security implications.

Wireshark was also used to capture and examine network traffic generated during the scanning process. This provided an opportunity to observe TCP packets associated with the Nmap scan and gain a practical understanding of how network scanning activity appears at the packet level.

This task provided me with practical experience in **network reconnaissance, TCP port scanning, service identification, packet analysis and basic security risk assessment**. It also demonstrated the importance of identifying unnecessary or inadequately secured services to reduce the potential attack surface of devices within a network.

2. OBJECTIVES

The objectives of this task were to:

- Identify the local network range.
- Perform a TCP SYN scan using Nmap to discover active hosts and open TCP ports.
- Record the IP addresses and open ports identified during the scan.
- Determine the common services associated with the open ports.
- Research the purpose of the identified services.
- Assess potential security risks associated with exposed network services.
- Use Wireshark to observe and analyse network traffic generated during the scan.

3. TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Nmap 7.98	Network discovery and TCP port scanning
Wireshark	Packet capture and network traffic analysis

4. LOCAL NETWORK IDENTIFICATION

Network: **192.168.88.0/24**

The /24 CIDR notation corresponds to the subnet mask **255.255.255.0**.

Parameter	Value
IPv4 Address	192.168.88.xxx
Subnet Mask	255.255.255.0
Network Address	192.168.88.0
Broadcast Address	192.168.88.255
Usable Host Range	192.168.88.1 – 192.168.88.254
CIDR Notation	192.168.88.0/24

5. NMAP SCANNING METHODOLOGY

A TCP SYN scan was performed using Nmap to identify active hosts and open TCP ports within the local network.

Command used: **nmap -sS 192.168.88.0/24**

Scan Details -

Parameter	Result
Network Scanned	192.168.88.0/24
Scan Type	TCP SYN Scan (-sS)
IP Addresses Scanned	256
Hosts Up	5
Hosts with Open Ports	4
Scan Duration	11.46 seconds
Nmap Version	7.98

The raw Nmap scan output was saved as: **nmap_scan.txt**

6. NMAP SCAN SUMMARY

IP Address	Host Status	Open TCP Ports Detected
192.168.88.1	Up	5
192.168.88.2	Up	0
192.168.88.3	Up	6
192.168.88.4	Up	5
192.168.88.6	Up	4

Total open TCP port identified: 20

7. IDENTIFIED IP ADDRESSES, OPEN PORTS AND SERVICES DURING NMAP SCAN

IP Address	Port	Service
192.168.88.1	22	SSH
192.168.88.1	23	Telnet
192.168.88.1	53	DNS
192.168.88.1	80	HTTP
192.168.88.1	1900	UPnP
192.168.88.3	22	SSH
192.168.88.3	23	Telnet
192.168.88.3	53	DNS
192.168.88.3	80	HTTP
192.168.88.3	443	HTTPS
192.168.88.3	1900	UPnP
192.168.88.4	135	MSRPC
192.168.88.4	139	NetBIOS-SSN
192.168.88.4	445	Microsoft-DS / SMB
192.168.88.4	3306	MySQL
192.168.88.4	6000	X11
192.168.88.6	135	MSRPC
192.168.88.6	139	NetBIOS-SSN
192.168.88.6	445	Microsoft-DS / SMB
192.168.88.6	3306	MySQL

Note: Host 192.168.88.2 was identified as active, but no open TCP ports were detected among the 1,000 ports scanned.

8. SERVICE RESEARCH

Port	Service	Purpose
22	SSH (Secure Shell)	Provides a secure, encrypted connection for remote login, command execution and system administration.
23	Telnet	Provides remote command-line access to a device. Telnet transmits communication without encryption and is therefore considered a legacy remote-access protocol.
53	DNS (Domain Name System)	Used for DNS services that translate domain names into IP addresses and support other forms of name resolution.
80	HTTP (Hypertext Transfer Protocol)	Used to deliver web pages and other web resources between clients, such as browsers and web servers.
443	HTTPS (HTTP Secure)	Used for secure web communication. It provides HTTP communication protected by TLS encryption.
135	MSRPC (Microsoft Remote Procedure Call)	Used by Windows systems to enable applications and services to communicate with processes running on other systems or within the operating system.
139	NetBIOS-SSN (NetBIOS Session Service)	Provides session-level communication for legacy Windows networking and is commonly associated with older file and printer sharing mechanisms.
445	SMB / Microsoft-DS	Used by the Server Message Block protocol for Windows file and printer sharing, network resource access and related services.
1900	UPnP (Universal Plug and Play)	Used in local networks for discovering devices and network services automatically, allowing compatible devices to identify and communicate with one another.
3306	MySQL	The commonly used network port for MySQL database server connections, allowing applications and clients to communicate with a MySQL database.
6000	X11	Commonly associated with the X Window System and used for communication between X11 graphical applications and the display server.

9. IDENTIFICATION OF POTENTIAL SECURITY RISKS FROM OPEN PORTS

a) Port 22 – SSH

Affected Hosts: 192.168.88.1, 192.168.88.3

Potential Risks:

- Weak or compromised credentials could allow unauthorized remote access.
- Outdated SSH implementations may contain known security vulnerabilities.
- Unnecessary exposure of SSH increases the remote-access attack surface.
- Password-based authentication may be targeted by brute-force attacks.

Risk Level: Medium

Recommended Mitigation:

- Use strong authentication and preferably SSH keys.
- Keep SSH and the underlying operating system updated.
- Restrict SSH access to trusted hosts or administrative networks.
- Monitor authentication logs for repeated or suspicious login attempts.

b) Port 23 – Telnet

Affected Hosts: 192.168.88.1, 192.168.88.3

Potential Risks:

- Telnet does not encrypt credentials or other transmitted data.
- An attacker with suitable network access may intercept sensitive information.
- Compromised Telnet credentials could result in unauthorized remote access.
- Telnet is an outdated and insecure alternative to encrypted remote administration.

Risk Level: High

Recommended Mitigation:

- Disable Telnet if it is not required.
- Replace Telnet with SSH for remote administration.
- If Telnet is temporarily required, restrict access to a trusted management network.

c) Port 53 – DNS

Affected Hosts: 192.168.88.1, 192.168.88.3

Potential Risks:

- Improperly configured DNS services may disclose internal network information.
- Unnecessary exposure of DNS increases the network attack surface.
- Misconfigured services may accept unauthorized DNS queries.
- Outdated DNS software may contain known security vulnerabilities.

Risk Level: Medium

Recommended Mitigation:

- Restrict DNS queries to authorized clients where appropriate.
- Keep DNS software and device firmware updated.
- Review DNS configuration and access-control settings.
- Disable unnecessary DNS functionality.

d) Port 80 – HTTP

Affected Hosts: 192.168.88.1, 192.168.88.3

Potential Risks:

- HTTP communication is unencrypted and may expose sensitive information if transmitted through the service.
- Vulnerabilities in the web server or web application could potentially be exploited.
- Poorly secured administrative interfaces or default credentials may allow unauthorized access.
- Information exposed by the web service may assist further reconnaissance.

Risk Level: Medium

Recommended Mitigation:

- Use HTTPS for sensitive communication.
- Redirect HTTP traffic to HTTPS where appropriate.
- Remove unnecessary web services and administrative interfaces.
- Keep web servers, firmware, and applications updated.
- Use strong authentication and appropriate access controls.

e) Port 443 – HTTPS

Affected Host: 192.168.88.3

Potential Risks:

- Vulnerabilities may exist in the web server or application even when HTTPS is enabled.
- Weak or outdated TLS configurations may reduce security.
- Vulnerable web applications can still be exploited through an encrypted connection.

Risk Level: Low to Medium

Recommended Mitigation:

- Keep the web server and application updated.
- Use current and secure TLS configurations.
- Disable obsolete protocols and weak cryptographic settings.
- Apply secure authentication and authorization controls.

f) Port 1900 – UPnP/SSDP

Affected Hosts: 192.168.88.1, 192.168.88.3

Potential Risks:

- UPnP can automatically expose network services and devices.
- Misconfigured or vulnerable UPnP implementations may increase the attack surface.
- Device and service information may be disclosed through network discovery.
- Unnecessary UPnP functionality may create unwanted network exposure.

Risk Level: Medium to High

Recommended Mitigation:

- Disable UPnP if it is not required.
- Restrict UPnP/SSDP traffic to trusted network segments.
- Keep router and device firmware updated.
- Review automatically created port mappings and exposed services.

g) Port 135 – MSRPC

Affected Hosts: 192.168.88.4, 192.168.88.6

Potential Risks:

- RPC services can expose information about Windows services and system functionality.
- Unnecessary exposure increases the attack surface of Windows systems.
- Vulnerable or misconfigured RPC services may potentially be exploited for unauthorized access.

Risk Level: Medium to High

Recommended Mitigation:

- Restrict access to port 135 using firewall rules.
- Allow RPC access only from trusted systems where required.
- Keep Windows systems fully patched.
- Disable unnecessary services.

h) Port 139 – NetBIOS

Affected Hosts: 192.168.88.4, 192.168.88.6

Potential Risks:

- NetBIOS may expose information about Windows systems and shared resources.
- Weak sharing permissions may result in unauthorized access to network resources.
- The service increases the attack surface of Windows systems.

Risk Level: Medium to High

Recommended Mitigation:

- Disable NetBIOS over TCP/IP where it is not required.
- Restrict port 139 using firewall rules.
- Review network-share permissions.
- Keep systems patched and securely configured.

i) Port 445 – SMB

Affected Hosts: 192.168.88.4, 192.168.88.6

Potential Risks:

- Unauthorized users may gain access to shared files and network resources if permissions are poorly configured.
- Weak authentication or excessive permissions may expose sensitive information.
- Vulnerable SMB implementations can present significant security risks.
- SMB exposure can facilitate lateral movement if a system is compromised.

Risk Level: High

Recommended Mitigation:

- Restrict SMB access using firewalls and network segmentation.
- Disable SMBv1 if it is enabled and not required.
- Use strong authentication and least-privilege permissions.
- Keep Windows systems and SMB components patched.
- Disable unnecessary file and printer sharing.

j) Port 3306 – MySQL

Affected Hosts: 192.168.88.4, 192.168.88.6

Potential Risks:

- Network-accessible database services may expose sensitive information if access controls are weak.
- Weak database credentials could result in unauthorized database access.
- Outdated MySQL installations may contain known security vulnerabilities.
- Unrestricted database access increases the attack surface.

Risk Level: High

Recommended Mitigation:

- Restrict MySQL access to authorized application servers and administrators.
- Avoid exposing the database to untrusted networks.
- Use strong, unique database credentials.
- Keep MySQL updated.
- Apply least-privilege permissions to database accounts.
- Enable appropriate logging and monitoring.

k) Port 6000 – X11

Affected Host: 192.168.88.4

Potential Risks:

- An exposed X11 service may allow unauthorized access to graphical sessions if authentication and access controls are weak.
- X11 may expose sensitive information associated with graphical sessions.
- An unnecessary network-based X11 service increases the attack surface.

Risk Level: High if exposed beyond trusted systems

Recommended Mitigation:

- Disable network-based X11 access if it is not required.
- Restrict access using firewall rules.
- Use secure mechanisms such as SSH tunnelling for remote graphical administration where appropriate.
- Apply proper X11 authentication and access controls.

10. OVERALL RISK ASSESSMENT

The scan indicates that several services may increase the network's attack surface. The highest-priority findings are:

Priority	Port	Service	Affected Hosts	Main Security Concern
1	23	Telnet	192.168.88.1, 192.168.88.3	Unencrypted remote access
2	445	SMB	192.168.88.4, 192.168.88.6	Unauthorized file/resource access and potential lateral movement
3	3306	MySQL	192.168.88.4, 192.168.88.6	Unauthorized database access
4	6000	X11	192.168.88.4	Potential exposure of graphical sessions
5	1900	UPnP	192.168.88.1, 192.168.88.3	Unnecessary device/service exposure

11. RECOMMENDED SECURITY ACTIONS

Based on the scan results, the following actions are recommended:

- Disable unnecessary services and close unused ports.
- Replace Telnet with SSH for secure remote administration.
- Restrict SMB, MSRPC, and NetBIOS access using firewall rules and network segmentation.
- Restrict MySQL access to authorized application servers and administrators.
- Disable unnecessary UPnP functionality.
- Disable or restrict X11 network access if it is not required.
- Keep operating systems, firmware and network services updated.
- Use strong authentication and least-privilege access controls.
- Apply network segmentation to limit unnecessary communication between systems.
- Monitor system and authentication logs for suspicious activity.
- Perform a follow-up vulnerability assessment after remediation.

12. ANALYSIS OF PACKET CAPTURE

A packet capture was performed using **Wireshark** on the local network before conducting the Nmap scan.

Capture File: **Packet_capture.pcapng**

The packet capture was reviewed to observe:

- Source and destination IP addresses.
- Network protocols present in the captured traffic.
- Communication between devices on the local network.
- DNS, TCP, and UDP traffic.
- Broadcast and multicast traffic generated by network devices.

This analysis provided an initial view of network activity before performing the Nmap port scan.

13. CONCLUSION

This task helped me develop a practical understanding of **basic network reconnaissance and port scanning** using Nmap. I learned how to identify the local IP range, perform a TCP SYN scan, detect active devices and open ports and understand the services associated with those ports. By analysing the result, it also helped me recognize how exposed services can increase network security risks. Using Wireshark provided additional exposure to observing network traffic and communication.

Overall, this task strengthened my understanding of **IP ranges, ports, network services, network exposure and basic security assessment techniques**.